

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Summer 2024

Course: B. Tech.

Semester: VI

Branch: Computer Engineering / Computer Science and Engineering

Subject Code & Name: BTHM605A_Y23 Development Engineering

Max Marks: 60

Date: 25/06/2024

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Mark

Q. 1 Solve Any Two of the following.

- A)** What is sustainable development? How does development engineering integrate with sustainable development goals? **6**
- B)** Define development engineering with the following perspectives: **6**
1. Problem-Solving Process
 2. Economic Development
 3. Technological Empowerment
- C)** Enlist and explain the guiding principles of economic development. **6**

Q.2 Solve Any Two of the following.

- A)** What are the prime responsibilities of a development engineer who intends to solve societal problems? **6**
- B)** Identify the major issues in the economic development of India and how development engineering can address them. **6**
- C)** Define the poverty line. Which poverty measure will help us to find the causes and consequences of poverty? **6**

Q. 3 Solve Any Two of the following.

- A)** Which guiding principle of economic development will follow the secular perspective? **6**
- B)** Explain why incorporating social justice in engineering is crucial for sustainable development. **6**
- C)** Explain the Amartya Sen's idea of social justice. **6**

Q.4 Solve Any Two of the following.

- A) Enlist and explain any three emerging technologies and their contribution to sustainable development. 6
- B) How do artificial intelligence and machine learning have a positive impact on the health sector? 6
- C) How do technological advancements contribute to a quality education? 6

Q. 5 Solve Any Two of the following.

- A) Explain the applications of blockchain technology for societal development. 6
- B) Discuss the importance of teamwork and project management in the context of community development projects. 6
- C) Explain the role of artificial intelligence for humanitarian purposes with suitable examples. 6

***** End *****